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CMSC 25422 Project Proposal

Project title

Exploring Beyond nPrint: Assessing the Impact of Engineered Features on Network Flow Classification

Project summary

What is the problem, and why is it important?

The main problem that the project is dealing with is Network traffic classification, which is the identification of what application or service generates a flow. Network traffic classification is central to network security and management, powering tools like intrusion detection and QoS enforcement. Standard approaches like nPrint turn packet headers into ML-ready features, but it's unclear whether additional engineering features can improve on that baseline. This project matters because identifying which packet-level signals actually drive classification accuracy can make network monitoring tools more effective and efficient.

Data

What data will you be using for the project?

We will use the provided compressed packet capture file (traffic.pcapng.gz), which contains 529,000 network packets. We will group these packets into network flows before extracting features and training our models.

Dataset Link: [traffic.pcapng.gz](https://www.dropbox.com/s/4833333333333333/traffic.pcapng.gz?dl=1)

Machine learning

What models do you expect to try?

Our main model will be LightGBM. We plan to use techniques such as sample weighting and adjust model settings to account for classes with different numbers of examples. We will also try to experiment with XGBoost and ExtraTrees. We will then compare their individual performance and test whether averaging the predictions will improve the results or not.

Evaluation

How will you evaluate your models?

We will mainly use balanced accuracy to evaluate our models because it gives each class equal importance, even when some classes may have fewer examples than others. We will also use

Macro F1 to evaluate how well the model handles precision and recall across all classes. Then, we will also compare the standard nPrint model with models that included our new features that we implemented to see whether the additional features improve classification performance.

Learning objective

What are you expecting to learn from your project?

We hope to get more experience working with PCAPs, learn how to organize raw packets captured into flows, extract important features from the packets, and get more familiar with different machine learning models.